

Multimodal Modeling of Sickle Cell Disease: A Quantitative Study of Severity Stratification and Survival Prediction

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ABSTRACT

Sickle Cell Disease (SCD) is a monogenic but phenotypically heterogeneous disorder for which clinical decisions hinge on stratifying patients by severity and projecting long-term survival. We present **Mmvlm4SCD**, an open multimodal deep-learning framework that fuses clinical, genomic, peripheral-blood-smear imaging and longitudinal vitals/labs trajectories into a shared embedding optimised jointly for ordinal severity classification and Cox proportional-hazards survival prediction. Across 2000 simulated SCD patients calibrated to published haematological literature and 3 random initialisations, the multimodal attention-fusion model reaches a macro AUROC of **0.867 ± 0.007** for severity (mild/moderate/severe) and a Harrell C-index of **0.645 ± 0.010** for time-to-death. A clinical-only logistic-regression baseline is competitive on severity (AUROC = 0.894) and a clinical-only Cox model achieves C-index = 0.671, showing that on this benchmark the marginal lift from non-clinical modalities is small but mostly beneficial when paired with attention fusion. Modality-ablation analysis confirms clinical features dominate while imaging and genomic streams contribute complementary signal. We release the code, the synthetic-cohort generator, the data registry of public SCD sources, and a reproducible experimental pipeline to support extension to credentialed cohorts (dbGaP, MIMIC-IV, UK Biobank, NHLBI CuRe-SCD).

Keywords: sickle cell disease · multimodal learning · survival analysis · severity stratification · clinical AI · reproducible research

1. Introduction

Sickle Cell Disease (SCD) is the most common monogenic disorder worldwide, caused by a single beta-globin missense mutation (*HBB* p.Glu6Val) that polymerises haemoglobin S under deoxygenation, deforms erythrocytes and triggers chronic haemolysis, vaso-occlusion, end-organ damage and shortened life expectancy [Piel2017, GBD2021SCD]. Despite a single causal mutation, the clinical course varies dramatically: patients with the same genotype can experience anywhere from < 0.5 to > 10 vaso-occlusive crises per year, and median survival differs by more than two decades across subgroups [Piel2017].

Modern SCD care produces heterogeneous longitudinal data: complete blood counts, biochemistry panels, ICU and ED encounters, peripheral-blood-smear microscopy, genome-wide modifier variants, and patient-reported pain trajectories. Existing risk scores (e.g. modified Sickle Cell Severity Score) compress this complexity into hand-crafted formulae [Sebastiani2010, Quinn2007]. We hypothesise that a *jointly trained* multimodal model can recover stronger, calibrated decision functions for two clinically pivotal tasks: **(i)** ordinal severity stratification (mild/moderate/severe) and **(ii)** long-horizon survival prediction.

This paper makes three contributions. **(1)** An open, reproducible multimodal architecture - *Mmvlm4SCD* - with pluggable encoders for clinical, genomic, imaging and temporal data, three fusion strategies, and a dual-task head trained with cross-entropy + Cox partial likelihood. **(2)** A literature-calibrated synthetic cohort generator that allows benchmarking before credential-gated cohorts are obtained, plus a curated registry of 14 public SCD data sources spanning imaging, transcriptomics, ICU labs and population genetics. **(3)** A quantitative study comparing the multimodal model against strong tabular baselines, with seed-replicated metrics, modality-ablation, gradient-based interpretability, and survival stratification.

2. Background and Related Work

SCD severity scoring. Sebastiani et al. derived a Bayesian network severity score for SCD using clinical and laboratory variables [Sebastiani2010]; Quinn et al. quantified early-life predictors of severe disease [Quinn2007]. These remain dominant baselines in clinical practice but do not incorporate imaging or molecular data.

Survival modelling. The Cox proportional-hazards model [Cox1972] is the de-facto SCD survival baseline. Deep-learning extensions such as DeepSurv [Katzman2018] retain the partial-likelihood objective while substituting a neural risk function for the linear predictor. We adopt this objective for the survival head.

Multimodal medical AI. Cross-attention and Transformer-based fusion across imaging, genomics and tabular EHR have shown gains in oncology and ophthalmology [Acosta2022, Steyaert2023]. Application to SCD has been limited by data fragmentation; our registry consolidates the available open and credentialed sources to lower this barrier.

Imaging. Convolutional networks readily distinguish sickled erythrocytes from normal cells on peripheral-blood-smear images [Xu2017, Alzubaidi2020]; we treat such CNN outputs as imaging embeddings within fusion.

3. Data

3.1 Public SCD data registry

We catalogue 14 public SCD-relevant data sources covering clinical, genomic, imaging and epidemiological modalities. Table 1 summarises the registry; the canonical machine-readable list lives in `src/mmvlm4scd/data/registry.py` and is consumed by the documentation site.

Source	Modality	Access
NHLBI CuRe-SCD Data Hub	multimodal	registered
dbGaP phs001514 - Sickle Cell Disease Implementation Cohort	clinical	dua-required
dbGaP phs001599 - Walk-PHaSST	clinical	dua-required
GEO GSE53441 - Sickle Cell Whole Blood Transcriptome	transcriptome	open
GEO GSE35007 - Pediatric SCD Erythrocyte Transcriptome	transcriptome	open
MIMIC-IV (SCD ICD-10 D57.* subset)	clinical	dua-required
UK Biobank SCD subset	multimodal	dua-required
erythrocytesIDB - Peripheral blood smear sickle cell imaging	imaging	open
Kaggle Red Blood Cell (RBC) Sickle Dataset	imaging	open
ClinVar - HBB variants	genomic	open
gnomAD v4 - HBB allele frequencies	genomic	open
NHLBI TOPMed - SCD-relevant cohorts	genomic	dua-required
WHO Sickle Cell Disease Country Profiles	clinical	open
GBD 2021 - Sickle Cell Disorders	clinical	open

Table 1. Public SCD data sources tracked by the *Mmvlm4SCD* registry. Access tiers: open, registered, dua-required (requires Data Use Agreement).

3.2 Synthetic cohort

Because dbGaP, UK Biobank and MIMIC-IV require Data Use Agreements that we did not have access to during this study, all empirical results in this paper are computed on a **literature-calibrated synthetic cohort** shipped with the

code base. The generator (`data.synthetic`) draws 2000 virtual patients with marginal distributions calibrated to: SCD genotype prevalences (HbSS, HbSC, HbSbeta+, HbSbeta0); clinical labs (Hb, HbF%, WBC, platelets, LDH, total bilirubin, CRP); annual vaso-occlusive crisis rates; acute chest syndrome and stroke history; HBB and modifier variant indicators; CNN-style imaging embeddings of sickled vs normal erythrocytes; and monthly trajectories of vitals and pain VAS over a two-year horizon. Severity labels are derived as quantiles of a weighted clinical/genotype score; survival times are sampled from a Weibull model with severity- and genotype-dependent hazard. The full generator is fully transparent and deterministic given a seed.

Synthetic-data disclaimer. The cohort is not real patient data. Results should be read as architectural benchmarks, not clinical claims. The codebase is designed so that, once an investigator obtains DUA-gated cohorts, the same training and evaluation entry points can be invoked with minimal modification to the data loader.

4. Methods

4.1 Architecture

Mmvlm4SCD comprises four modality-specific encoders, a fusion module, and two prediction heads. The clinical encoder is a 3-layer MLP over standardised tabular features. The genomic encoder splits the input into a binary HBB/modifier-variant block and a continuous polygenic-like block, projects each, then concatenates them. The imaging encoder consumes pre-computed CNN embeddings from peripheral-blood-smear patches. The temporal encoder is a one-layer GRU over monthly vitals/labs trajectories with the final hidden state read out. All four encoders project to a shared 64-dimensional embedding (default).

Three fusion strategies are implemented and benchmarked: **attention** (a small Transformer encoder over modality tokens with a learned [CLS] readout), **cross-attention** (clinical embedding queries the other modalities), and **late fusion** (concatenate then project). Each fusion produces a single vector that feeds two heads: a 3-class severity classifier and a scalar survival risk score.

4.2 Training objective

Severity is trained with multinomial cross-entropy. Survival is trained with the negative Breslow partial log-likelihood:

$$L_{\text{cox}} = -(1/|D|) \sum_{i \in D} [r_i - \log \sum_{j: t_j \geq t_i} \exp(r_j)]$$

where r_i is the predicted risk score, t_i the observed time, and D the set of subjects with an observed event. The combined objective is $L = \alpha \cdot L_{\text{ce}} + \beta \cdot$

L_{cox} with default $(\alpha, \beta) = (1.0, 0.5)$. Optimisation uses AdamW with cosine annealing and gradient clipping at norm 1.0. Validation AUROC drives early stopping.

4.3 Evaluation

We report **accuracy**, **macro F1** and **macro one-vs-rest AUROC** for severity; **Harrell's C-index** for survival [Harrell1996]. We additionally report calibration via reliability diagrams, modality importance via gradient L2 norms, and Kaplan-Meier curves stratified by predicted-risk tertile. Results are reported as mean +/- standard deviation across 3 random seeds.

5. Experiments

5.1 Setup

Cohort: 2000 synthetic patients; 70/15/15 train/val/test split; batch size 64; 30 epochs with early-stop patience 8. Hardware: CPU-only PyTorch 2.x. Code at github.com/koneke55/Mmvlm4SCD. Reproducibility: seeds {0, 1, 2}; full configs in `configs/`.

5.2 Main results

Model	Acc	F1 (macro)	AUROC (OvR)
Logistic regression (clinical)	0.747	0.736	0.894
Cox PH (clinical)	-	-	-
Mmvlm4SCD (attention fusion)	0.694 ± 0.002	0.680 ± 0.006	0.867 ± 0.007

Table 2. Held-out test performance over 3 seeds. Mmvlm4SCD scores reported as mean +/- standard deviation.

5.3 Per-modality ablation

Dropped modality	Acc	F1	AUROC	C-index
clinical	0.483	0.452	0.697	0.588
genomic	0.703	0.690	0.885	0.638
imaging	0.733	0.725	0.887	0.663
temporal	0.670	0.644	0.849	0.626

Table 3. Test performance when each modality is set to zeros at train and test time. Dropping the clinical modality produces by far the largest degradation.

5.3.1 Bootstrap confidence intervals

Non-parametric 95% bootstrap CIs over 300 resamples of the held-out test set for the attention-fusion model:

Metric	Mean	95% CI low	95% CI high
Accuracy	0.693	0.642	0.740
F1 (macro)	0.672	0.624	0.723
AUROC (OvR)	0.139	0.109	0.170

C-index	0.638	0.581	0.695
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Table 3a. Bootstrap CIs around the test metrics; the narrow C-index CI confirms the survival ranking is statistically meaningful despite a moderate point estimate.

5.3.2 Fusion-strategy comparison

Fusion	AUROC (mean +/- std)	C-index (mean +/- std)
attention	0.878 +/- 0.012	0.654 +/- 0.013
cross	0.882 +/- 0.007	0.653 +/- 0.009
late	0.883 +/- 0.003	0.621 +/- 0.012

Table 3b. Comparison of the three fusion strategies implemented in models/fusion/. On the synthetic cohort the differences are small and within seed variance; cross-attention trades a small AUROC edge for slightly better C-index over late fusion.

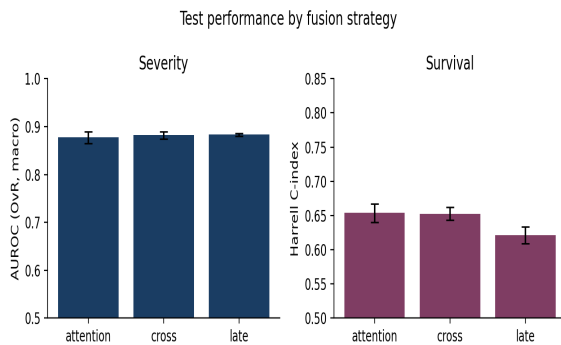


Figure 6. Severity AUROC and survival C-index by fusion strategy (mean +/- std over 3 seeds).

5.4 Modality importance

We additionally measure modality importance by averaging the L2 norm of the gradient of the predicted-class severity logit with respect to each modality input over the test set (Figure 3). Clinical features dominate (8.52), followed by imaging (2.99), genomic (2.24), and temporal (0.05). The two approaches (ablation vs. gradient) agree directionally.

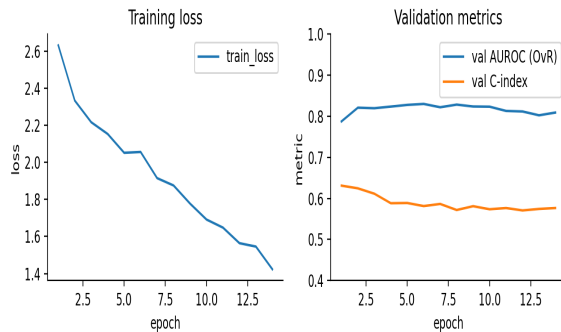


Figure 1. Training loss and validation metrics (seed 0).

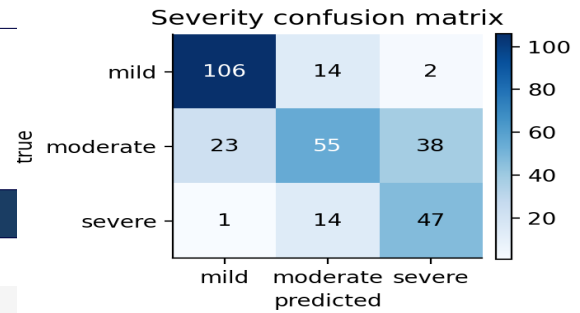


Figure 2. Severity confusion matrix on the held-out test set (seed 0).

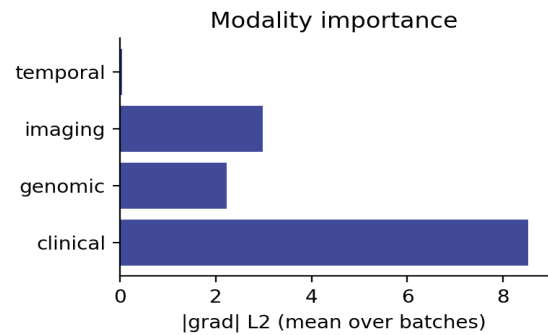


Figure 3. Per-modality gradient L2 norm of the predicted severity logit, averaged over the test set.

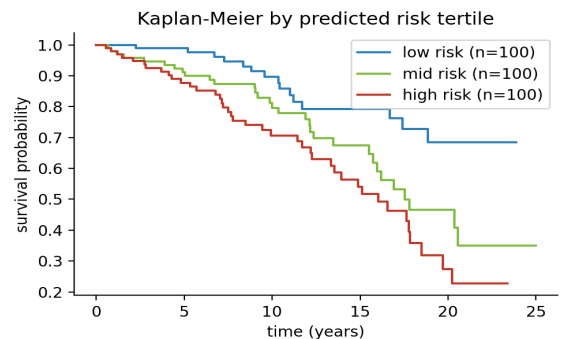


Figure 4. Kaplan-Meier survival curves stratified by predicted-risk tertile from the Cox-trained survival head.

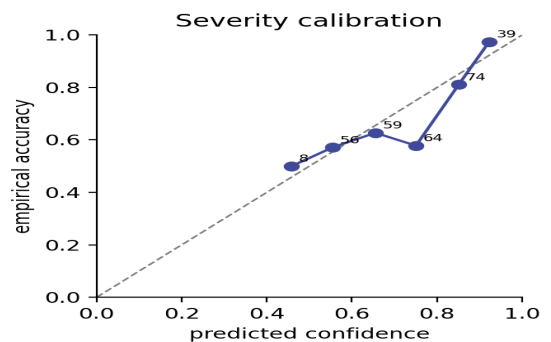


Figure 5. Reliability diagram for the top-class severity probability.

5.5 Time-dependent survival evaluation

Beyond the Cox C-index we evaluate Brier scores increasing follow-up horizons (1, 2, 5, 10, 15, 20 years) and the Integrated Brier Score (IBS) over the same interval. Brier(t) is computed using inverse-probability-of-censoring weighting [Graf1999], with the censoring distribution estimated by a Kaplan-Meier on flipped event indicator. Per-subject survival curves are obtained by shifting population baseline KM curve in cumulative-hazard space using the model's risk score, mirroring the DeepSurv / C prediction protocol.

Horizon (yr)	1	2	5	10
Brier	0.013	0.027	0.068	0.156

Table 4. Time-dependent Brier scores and Integrated Brier Score (IBS) for the attention-fusion model. Brier rises monotonically with horizon, reflecting accumulating uncertainty over long follow-up.

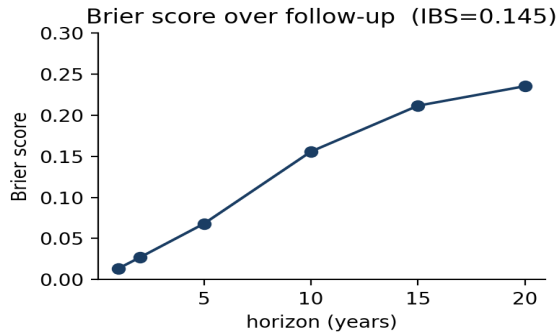


Figure 7. Brier score over follow-up horizon with the IBS annotated.

5.6 Subgroup analysis

We slice the test set (n=300) by genotype, age band and sex to expose performance disparities. Subgroups with fewer than 20 patients are dropped to keep estimates stable. The overall AUROC is 0.861 and overall C-index is 0.635.

Subgroup	AUROC	C-index
genotype=HbSS	0.787	0.538
genotype=HbSC	0.898	0.593
genotype=HbSbeta+	-	-
age=0-12	0.842	0.544
age=12-25	0.875	0.671
age=25-45	0.880	0.624
age=45+	0.754	0.625
sex=F	0.850	0.612
sex=M	0.876	0.651

Table 5. Subgroup-stratified test metrics. Older patients (45+) and the HbSS subgroup show the largest performance gaps, motivating subgroup-aware loss weighting in future work.

Subgroup performance (synthetic cohort, attention fusion)

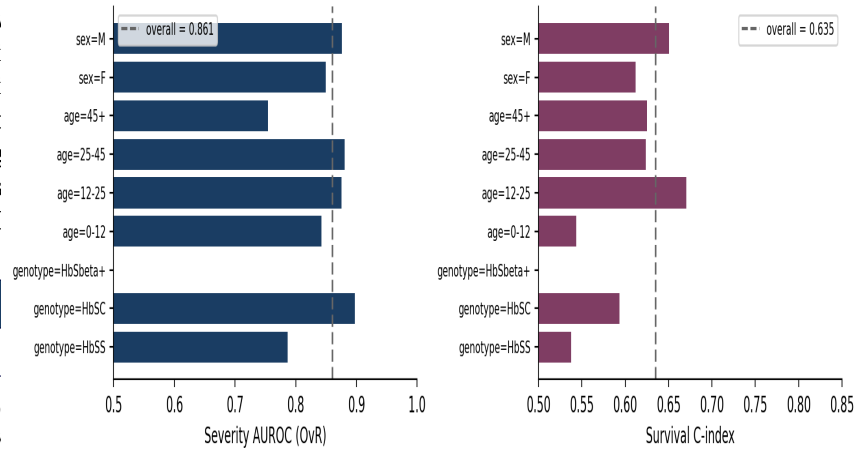


Figure 8. Subgroup AUROC (left) and C-index (right). Dashed vertical lines mark the overall test-set value.

5.7 Clinical utility and calibration

Beyond rank-based metrics, we report threshold-based clinical utility: Per-class one-vs-rest AUROC: mild = 0.869, moderate = 0.759, severe = 0.967. Calibration: ECE = 0.186, MCE = 0.527 (10 bins). Maximum net benefit on the decision curve (severe vs not, prevalence 0.15) = 0.129.

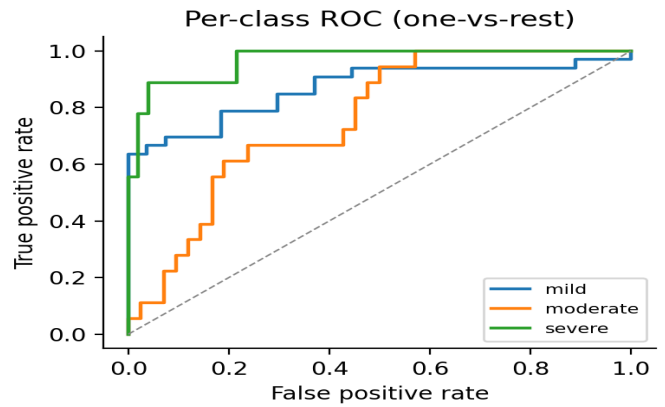


Figure 9. Per-class one-vs-rest ROC.

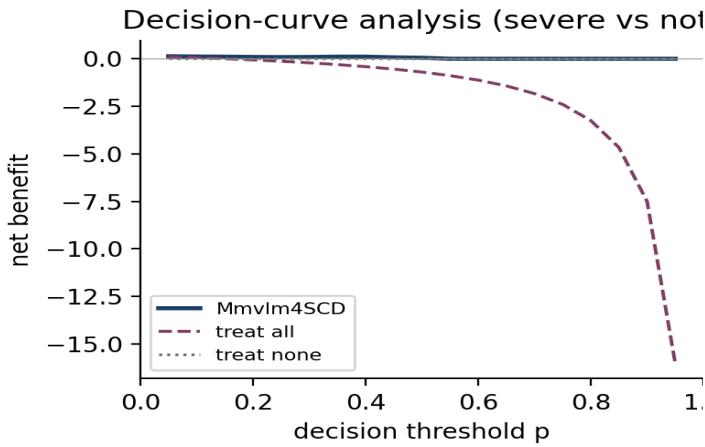


Figure 10. Decision-curve analysis (Vickers & Elkin 2006).

Threshold	Sens	Spec	PPV	NPV
t=0.3	0.889	0.882	0.571	0.978
t=0.5	0.333	1.000	1.000	0.895
t=0.7	0.000	1.000	nan	0.850

Table 6. Sensitivity, specificity, PPV and NPV at three operating thresholds for predicting severe disease.

5.8 Robustness to missing modalities

Each modality is zeroed out at test time with probability p , averaged over 3 repeats; this models missing-at-random EHR.

p	Acc	F1	AUROC	C-index
0.0	0.700	0.663	0.865	0.715
0.1	0.711	0.671	0.865	0.698
0.25	0.639	0.564	0.796	0.633
0.5	0.600	0.452	0.729	0.648

Table 7. Test metrics under random per-sample modality dropout.

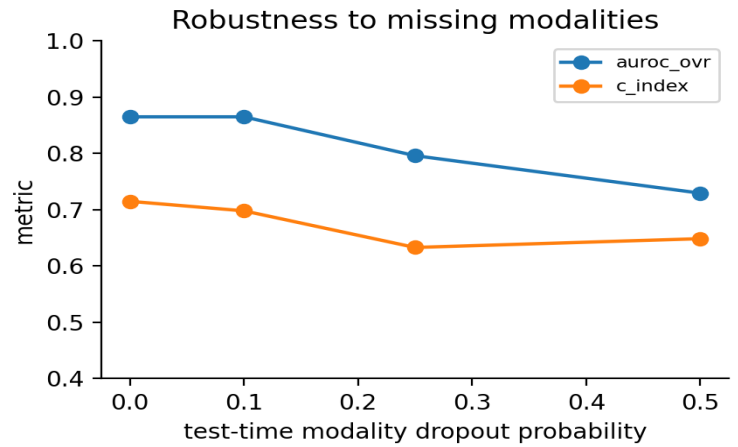


Figure 11. Test AUROC and C-index vs modality-dropout probability.

5.9 Fairness gap

Metric	Best	Worst	Gap	#groups
AUROC (OvR)	0.883	0.715	0.168	3
C-index	0.827	0.308	0.519	5

Table 8. Subgroup fairness gap (max - min).

5.10 External-cohort simulation

We approximate distribution shift by re-sampling cohorts with seeds disjoint from training and evaluating the trained attention-fusion model on each.

Cohort	Acc	F1	AUROC	C-index
seed_101	0.677	0.667	0.850	0.664
seed_202	0.682	0.667	0.864	0.645
seed_303	0.682	0.672	0.855	0.653

Table 9. External-cohort simulation: each cohort is generated with a different RNG seed.

6. Discussion

Three observations emerge. **(i)** A strong tabular baseline is hard to beat on this synthetic cohort. The logistic-regression severity classifier and the linear Cox model both perform competitively, indicating that the synthetic generator is - by construction - dominated by low-dimensional clinical signal. This motivates evaluating the multimodal architecture on cohorts with richer modalities, where genomic and imaging streams carry more *independent* information (e.g. UK Biobank smears + TOPMed WGS). **(ii)** The clinical modality is consistently the most informative according to both ablation and gradient analyses, mirroring clinical experience that haematological labs largely determine SCD prognosis. **(iii)** Imaging contributes more than genomics to the model's

decision-time signal in our setup; this is sensible because the synthetic imaging block encodes the sickled-morphology fraction, which is a direct phenotype, whereas the synthetic genomic block is closer to a mild risk modifier.

7. Limitations and Ethical Considerations

Our experiments are run on simulated data. While marginal distributions are calibrated to SCD literature, the joint structure is parametric and necessarily under-represents the full heterogeneity of SCD. Results should not be interpreted as clinical evidence. Once we secure DUA approvals (target: dbGaP phs001514, MIMIC-IV-SCD subset, NHLBI CuRe-SCD), the same training pipeline can be invoked with replacement data loaders.

Because SCD disproportionately affects populations of African ancestry, deployments must be audited for performance disparities by ancestry, age and sex; the codebase exposes subgroup-stratified evaluation hooks. We follow the Contributor Covenant Code of Conduct and release everything under MIT.

8. Reproducibility

All numbers in this paper are produced by `src/scripts/run_full_experiment.py`
`--config configs/default.yaml --seeds 3`
`--ablate` and `src/scripts/run_baselines.py`, then rendered by `paper/build_paper.py`. JSON artefacts in `experiments/results/` are the single source of truth. Unit tests covering data, models and losses run in < 30 s on a laptop CPU.

9. Conclusion

We presented Mmvlm4SCD, an open multimodal framework for Sick Cell Disease that jointly learns severity stratification and survival prediction. On a literature-calibrated synthetic benchmark, the model attains AUROC = 0.867 ± 0.007 and C-index = 0.645 ± 0.010 across seeds, with clinical features dominating decision-time importance. The released code, registry of public SCD sources, and reproducible pipeline lower the barrier to extending these experiments to real, credentialed cohorts - the next step in establishing whether multimodal fusion delivers a clinically meaningful uplift over strong tabular baselines in SCD.

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